JOHNATHAN AHDOUT

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Education

Princeton University *Present – May 2029*

B.A. in Mathematics, Minors in Computer Science & Optimisation and Quantitative Decision Sciences

- Cumulative GPA: 3.79.
- Relevant Coursework: Multivariable Calculus (MAT 201), Fundamentals of Statistics (ORF 245), Data Structures and Algorithms (COS 226), Single Variable Analysis with an Introduction to Proofs (MAT 215), Honours Linear Algebra (MAT 217), Probability Theory (MAT 385), Combinatorics (MAT 377), Algebra 1 (Abstract Algebra, MAT 345), Financial Investments and Derivatives (ECO 362).

John L. Miller Great Neck North High School *Sep 2021 – Jun 2025*

High School Diploma (Salutatorian)

- SAT: 1590 (800M, 790R&W); GPA: 97.8/100.
- Relevant Coursework: AP Chemistry (5), AP World History (5), AP Computer Science A + DSA (5), AP Biology (5), AP English Language and Composition (5), Senior Research Seminar, AP Statistics (5), AP Physics C: Mechanics (5), AP Physics C: Electricity & Magnetism (5), AP Calculus BC (5), Advanced Topics in Mathematics (Linear Algebra, Number Theory, Topology, Algebraic Geometry), AP Macroeconomics (5), AP Microeconomics (5).
- 2025 Departmental Awards: Overall Excellence in Mathematics; Overall Excellence in Computer Science.

Research & Professional Experience

Dickson Lab @ Michigan State University, ACRES REU *May 2026 – Present*

NSF Funded REU on Diffusion ML Modelling for Molecular Dynamics

- Selected for the ACRES REU (10 of 200+ applicants) to conduct research under Dr. Alex Dickson on conditional score-based diffusion models for energy-based 3D molecular generative inpainting.
- Extending the LEVI framework in PyTorch that couples SE(3)-equivariant graph neural networks with physics-based molecular dynamics to generate valid drug-like ligands within protein binding pockets.
- Optimising multi-GPU training and inference workflows on ICER's HPCC, and exploring equivariant diffusion network designs including attention-based pocket conditioning & adaptive step-size scheduling in the reverse SDE using RePaint/DiffLinker transferrable energy-based learning + inference.
- Developed **ACForce**, a novel graph-connectivity objective based on Laplacian graph theory that increased molecular connectivity by over 30% & by nearly 50% on benchmark drug-like molecule datasets while preserving molecular quality; continuing work to implement network flow & effective resistance methods to prevent overcompacted & valence-inconsistent molecule generation.
- Deliverables: poster & presentation at MSU Mid-SURE conference, rigorous LaTeX proofs of force/energy asymptotic loss analysis & novel Laplacian graph & Fiedler score connectivity for molecular cohesion, and anticipated co-authorship on a manuscript to be submitted to NeurIPS and ICLR, as well as first-author in a proof-based diffusion paper for submission to JCIM. First place presentation at ACRES Symposium.

Independent Research, Princeton University *Aug 2026 – Present*

Conditional Mutual Information for Point-in-Time Macro-Quantamental Screening

- Developed a research proposal applying conditional mutual information to screen point-in-time macroeconomic indicators from J.P. Morgan's Macrosynergy Quantamental System (JPMaQS) for predictive content beyond linear & existing information-theoretic measures. Working with Macrosynergy Quant Researchers & Princeton ORFE faculty to develop and publish.
- Derived the bias–variance rate of a k-nearest-neighbour conditional-mutual-information estimator & designed a cross-country panel-pooling procedure, with a poolability test, to relax its dimensionality limit.
- Proposed a diffusion-model-based estimator, built on recent score-based mutual-information methods, to admit high-dimensional text embeddings (FOMC statements, earnings-call transcripts) as screening indicators.

Princeton Physics (with Ph.D. Tian-Hua Yang) *Jan 2026 – Mar 2026*
Tensor Network Algorithms Directed Reading on Current Methods in Quantum Many-Body Dynamics

- Conducted literature review on non-Hermitian tensor-network DMRG and effective Liouvillian/non-normal operators in open systems to identify slow modes, metastability, and long-time relaxation structure.
- Extended/reproduced Julia workflows in ITensors.jl: biorthogonal left/right eigensystems, Krylov/Arnoldi spectral targeting, variational sweeps, adaptive SVD truncation, and residual/entanglement-based convergence diagnostics.

Brown Lab @ Vanderbilt University Department of Pharmacology *Jun 2023 – Jun 2025*
Protein Dynamics & Cheminformatics Research Intern

- Worked with Dr. Benjamin Brown on graph neural-network workflows over ∼2M small drug-like molecules; implemented PyTorch-based computational drug-design modelling with weekly technical presentations.
- Built protein-dynamics simulations with MCMC-informed sampling for dopamine partial-agonist affinity/potency modelling.
- Achieved 85% predictive accuracy via fine-tuning/transfer learning in metabolic-stability prediction settings.
- Regeneron STS 2025 Top 300 project: Contextualising Data-Sparse QSPR Property Prediction with Categorical Representations Built from Variational Autoencoders. Accepted to Princeton Undergraduate Research Journal 2026.

Yazawa Lab @ Icahn School of Medicine at Mount Sinai, New York, NY *Fall 2023 – Fall 2024*
Structural Biology and Immunology Research Intern

- Modelled SARS-CoV-2 E-protein pentameric structure and inhibitory peptide interfaces using molecular dynamics, geometric modelling, and energy minimisation workflows.
- Co-authored *PLOS Biology* publication: Developing inhibitory peptides against SARS-CoV-2 Envelope protein.
- Conducted conformational simulation studies on pyruvate/lactate biosensors to support structural validation.

Horowitz Lab @ University of Denver *Winter 2023 – Spring 2025*
Development & Design Intern (Foldit Education Mode)

- Collaborated on citizen-science protein-biophysics educational game development.
- Edited 100+ protein structure files and designed/implemented JavaScript educational game mechanics (sequence alignment and disulfide-bonding levels).
- Piloted and iterated activities with five AP Biology classes, refining puzzle scaffolding and instructional clarity from classroom feedback.

Science Squad & Columbia Undergraduate Research Journal *Feb 2023 – Sep 2024*
Research & Science Communication Intern

- Proposed a prime-number-based mutation detection algorithm for AlphaFold EvoFormer optimisation; finalist at Columbia University Science Journal Fair.
- Produced STEM communication content and research explainers through GameFormuLab YouTube channel scripting, editing, and digital publishing about physics concepts for young learners.

Mathey College Office Assistant, Princeton University Residential College *Sep 2025 – Present*
Office Assistant

- Manage front-desk operations for a 600+ student residential college, including scheduling, logistics, event support, and administrative coordination.
- Produce event communication assets and provide student-facing operational support, including leading programming for the Mathey Mooseletter and Peter S. Firestone Book Fund.

Teen TV Internship, North Shore Television *Fall 2022 – Spring 2023*
TV/Film Intern

- Trained across live-broadcast pipeline: camera operation, directing, TD panel work, audio/graphics, floor management, and green-screen compositing.
- Wrote segments/scripts for district-wide broadcast content including the weekly "5 o'Clock Live" segment.

Projects

Options Pricing & Volatility-Trading Research Toolkit (Python) *Jun – Aug 2026*

- Engineered three pricing engines (Black–Scholes, Monte Carlo, CRR binomial tree) behind a shared interface.
- Benchmarked Newton–Raphson, Brent, and Jaeckel/Halley IV solvers across 250 cases; Newton diverges on 14.4% of a realistic grid, motivating Jaeckel as the default.
- Derived a closed-form gamma/theta P&L identity from Itô's lemma and the Black–Scholes PDE, validated against a simulated self-financing delta-hedge.
- Implemented a 2-state Gaussian HMM from scratch (Baum–Welch EM, Viterbi) to detect volatility regimes and drive a gamma-scalping strategy.
- Built a 97-test suite (pytest) with regressions for bugs found during development, and generated a live SPY implied-vol surface exposing the smile absent from constant-volatility Black–Scholes.

GridSmart: Real-Time Grid Optimisation & Smart Home Scheduling for Equitable Energy
Princeton University Claude Builder Club Hackathon April 2026

- Built a real-time ERCOT grid optimisation dashboard judged by Princeton ORFE and COS faculty through video and in-person judging; web-app ingests hourly settlement point pricing across 8 Texas load zones, overlays fuel mix generation data to compute carbon intensity per hour, and runs a constrained greedy scheduling algorithm to optimally place household appliance runtimes within user-defined availability windows while staggering high-draw loads to prevent residential demand spikes (GitHub).
- Engineered end-to-end smart home automation by integrating Samsung SmartThings and Amazon Alexa device control APIs to programmatically trigger appliance cycles, toggle smart plugs, and pre-cool thermostats during renewable-surplus windows, transforming the scheduling engine's output from passive recommendations into autonomous execution across connected household devices with real-time confirmation and error handling.
- Designed an AI-powered data pipeline and equity analysis layer using Claude Haiku for zero-shot structured extraction of messy ERCOT HTML tables and CSVs into normalised JSON, with synthetic fallback generation calibrated to historical Texas wholesale pricing distributions by zone, feeding a geospatially accurate Leaflet-based frontend with 48-hour price and renewable forecasts, a tunable cost-versus-clean priority model, and a per-zone energy burden analysis quantifying the share of median household income spent on electricity to surface where community solar investment would have the highest social impact.

SPIRAL: Structured Physics-Informed Representation-Augmented Learning *Sep 2025 – Jan 2026*
Lead Research Developer Project Repository

- Built physics-informed multi-hazard forecasting pipeline fusing atmospheric gravity-wave features with real-time USGS/NOAA streams in PyTorch.
- Achieved $R^2 \approx 0.94$ and $\sim 54\%$ error reduction versus ARIMA and standard regression baselines.
- Deployed end-to-end inference and alerting stack (React/Next.js + Expo) with vectorised preprocessing and batch inference for continuous regional risk scoring.
- Presented poster and prototype at Princeton Research Day (Feb 2026).

Princeton Activities, Competitions & Technical Communities

Princeton Puzzles Club *Sep 2025 – Present*
Officer & Princeton Puzzlehunt Puzzle Writer/Tester

- Design and test Puzzlehunt-grade logic/meta puzzles (semaphores, research-based puzzles, etc.), including uniqueness checks, extraction-path construction, and calibrated solve difficulty.
- Playtested and designed 3 puzzles for Princeton Puzzlehunt 2026, which had 500+ attendees, real-time grading, bugfixing and running hint economy.

Princeton Game Theory Club *Sep 2025 – Present*
President, Previously: Events/Programming Director, Strategy Team Associate, Newsletter Editor

- Organise meetings and tournaments centred on strategic interaction, iterated games, and adversarial reasoning.

- Author structured game-prep materials for club distribution, including equilibrium framing, exploitability heuristics, and matchup-specific strategic adaptations.
- Write weekly newsletters advertising ongoing work in Nash equilibria, mechanism design, cooperative/competitive strategy, and iterated game dynamics for broader sponsor/department engagement.
- Wrote a LaTeX strategy guide for Fish covering mixed-strategy equilibria, payoff matrices, action paradoxes, and decision-tree optimisation; led prep for a school-wide tournament and won the inaugural Princeton GTC Fish competition.

Princeton Quantitative Traders *Sep 2025 – Present*
Member & Data Science Workshop Contributor

- Participate in weekly probability, statistical modelling, and quant-finance machine-learning workshops in Python; solve time-constrained expected-value optimisation and risk/reward decision problems.
- Build and evaluate trading heuristics for latency/spread modelling, Kelly optimisation, and risk-adjusted decision-making in mock execution settings.
- Collaborate on conditional-probability games, market-microstructure style reasoning exercises, and sponsored campus trading competitions (interview-style strategy rounds, trading games from quant firms like Jane Street, Citadel, Five Rings, etc).

Princeton Data Science Club (DataDev) *Sep 2025 – Present*
DataDev Team Member

- Build reproducible Python pipelines (NumPy, Pandas, scikit-learn) for feature engineering, cross-validation, and error diagnostics.
- Organise Datathon programming to expand education and participation in applied modelling workflows.
- Co-develop peer tutorials on regression, classification, clustering, Bayesian reasoning, and model evaluation, with emphasis on clean notebook standards, modular pipeline design, and object-oriented programming.

Princeton Quiz Bowl *Sep 2025 – Present*
Team A Competitor NAQT School Stats

- Compete in ACF Fall, ACF Winter, IQBT, and other collegiate tournaments; earned Top-10 placement at ACF Winter Regionals with Princeton A-Team.
- Specialise in computational science, mathematics, physics, chemistry, and modern biology packet categories.
- Write and edit practice questions/packet material; contribute to high-frequency packet-review training and tossup conversion drills.

Association for Computing Machinery (ACM), Princeton Chapter *Sep 2025 – Present*
Club Member and COSCON Competitor

- Attend technical talks/workshops in cryptography, complexity, and applied computing through Plaintext group; participate in collaborative algorithm/debugging rounds.
- Contributed to COSCON freshman-team algorithmic design and implementation tasks.

High School Leadership & Activities

TV North (TV Crew and Audio-Visual Club) *Fall 2021 – Spring 2025*
President / Co-President (3 years)

- Managed 10–20 person technical crews for live filming of districtwide events; handled equipment operations, lighting, audio engineering, and production timelines.
- Founded first GNNHS Film Festival; produced a multi-hour event, trailer, and full-length student film through a structured yearlong production cycle.
- Led weekly technical workshops on editing, screenwriting, and camera operations; supervised multi-camera productions for graduations and performances.

Science Olympiad Team *Fall 2021 – Spring 2025*

President & Captain (formerly Vice President/Co-Captain; Member)

- Member (2021–2023), Vice President & Co-Captain (2023–2024), President & Captain (2024–2025).
- Directed budgeting, logistics, scheduling, and peer education (chemistry, materials science, circuitry) across 20+ events serving 100+ students.
- Supported cross-club Physics Festival and research-education meetings as co-president across CS Club, Engineering Club, and Science Research Club for 3+ years, reaching 500+ students school-wide.
- Team ranked 6th overall at Nassau West Regionals (2024).

Science Research Club *Fall 2022 – Spring 2025*

Founding President

- Founded the school's first research-extension club; ran weekly meetings with slideshow curricula, lab preparation, and live demonstrations.
- Taught and executed wet-lab procedures including fungal crossing (Sordaria), PCR, gel electrophoresis, glass bending, microscopy, and chemical demonstrations (permanganate fireworks, non-Newtonian fluids, elephant toothpaste).
- Ran physics/engineering demonstrations including Van de Graaff generator experiments; guided students through gene observation and lab data collection workflows.

Computer Science Club *Fall 2022 – Spring 2025*

President

- Ran technical meetings and prepared officer slide decks; taught combinatorics, algorithms, Turing machines, quantum computing, data structures, and probability.
- Led computational chemistry and protein-modelling workshops using Chimera and AlphaFold.
- Taught introductory deep-learning modules with PyTorch + NumPy and coached student-led project design for DL research.

School Newspaper *Fall 2021 – Spring 2025*

Editor-in-Chief, Online Sector (formerly Associate Editor, Managing Editor)

- Associate Editor (2022–2023), Managing Editor (2023–2024), Editor-in-Chief (2024–2025).
- Wrote and edited student articles on science current events, politics, and public health.
- Led 20+ presentations on broadcast and digital journalism informed by Columbia University Journalism Conference training.
- Co-led launch of Broadcast and Digital Journalism social-media vertical with TV North, reaching 2000+ views.

Class Speaker / Student Council, Class of 2025 *Fall 2023 – Spring 2025*

Elected Class Representative

- Elected by Class of 2025; served on weekly committees and coordinated with administration to run Junior Prom, Pep Rally, fundraisers, and schoolwide programming.
- Invited external speakers (including politicians and authors) and represented the school at the Carle Place LI Leadership Symposium (1 of 2 delegates), BOCES Leadership Forum, and Unity Day 2024.

Columbia University Science Honours Program *Fall 2023 – Spring 2025*

Participant

- Completed weekly Saturday courses at Columbia Morningside: Python Healthcare Applications & Tools (Fall 2023), Quantum Physics and Relativity (Spring 2024), Advanced Topology (Fall 2024), Quantum Computing (Spring 2025).

Science Research Project Seminar (Science Fair) *Fall 2022 – Spring 2025*

Research Student

- Conducted multi-year research in thermodynamics, structural biology, and protein conformational energetics; presented at 6+ fairs (LISEF, LISC, NYSSEF, JSHS, WAC, SAAWA).
- Built ML models for Gibbs free-energy ΔG prediction across enzyme functional groups, integrating statistical

mechanics with protein-structure feature engineering.

- Developed computer-vision workflows (Python/OpenCV) modelling *D. melanogaster* magnetoreception to evaluate previously disputed findings.

Quiz Bowl (High School) *2021 – 2025*

Varsity/JV Competitor

- 2x Varsity second-place and champion placements in Long Island West formats; 2x JV champion; qualified for nationals in 2024 and 2025.

Teaching, Outreach & Service

STEM Tutor through Key Club and Tutor Tree *Fall 2021 – Sep 2024*

Volunteer and Paid Tutor

- Tutored Geometry Honours and Earth Science weekly (2 hrs/week, 24 weeks/year).
- Paid tutor for AP Biology, AP Chemistry, Algebra II/Trig, and SAT prep at $50/hr (3–5 hrs/week, ~40 hrs/year).
- Created chemistry/physics/history review guides for 60-student schoolwide study groups; held exam review sessions and AP Physics tutoring through synagogue community.

FunDay Monday Volunteer *Summer 2022*

Community Service

- Set up event infrastructure, supported food distribution, recorded event media, and coached pickleball for seniors.
- Coordinated with hospitals, artists, and local officials during Nassau County Department of Services events; 60 hours logged.

Courses & Certifications

Certifications: J.P. Morgan Quantitative Research Job Simulation via Forage

Introduction to Bioinformatics (Bar-Ilan University, edX) *Summer 2022*

College Course

- Studied pairwise alignment, biological assays, systems biology, and ML methods for genomic/proteomic structural prediction.
- Implemented dynamic programming for pairwise sequence alignment with memoisation and tabulation.

Skills

- **Technical:** Python, Java, R, PyTorch, NumPy, Pandas, scikit-learn, Matplotlib, LaTeX, Excel, PowerPoint.
- **Language:** Advanced Spanish (speaker, reader, writer).

Honours, Awards & Distinctions

- Regeneron Science Talent Search 2025 – Top 300 National Scholar.
- National Merit $2500 Scholarship Recipient (2025).
- U.S. Presidential Scholar Semifinalist (2025).
- Coca-Cola Scholarship Semifinalist (2025).
- Hagan Scholarship Finalist (2025).
- AP Scholar with Distinction (2024, 2025).
- Winner, Inaugural Princeton Game Theory Club Fish Tournament.
- Great Neck North HS Departmental Awards: Overall Excellence in Mathematics; Overall Excellence in Computer Science.
- Alan L. Gleitsman Outstanding Graduate Award ($1500); Frederick Duclos Barstow Award ($500).
- LI Science Congress Distinguished Project + Highest Honours in Biomedical Sciences; NY State Science Congress competitor.

- Science Olympiad medals in Cell Biology, Optics, and Codebusters.
- IAC National Science Bee Nationals Qualifier (2024, 2025): 15th nationally (2024), 4th nationally (2025).
- International Environmental Science Olympiad (IESO) Qualifier (2024).